

NAT Gateway

Outbound-Only Internet Access for Private Subnets

Interview Preparation Guide

Problem & motivation - source NAT internals - port allocation - packet flow - failure modes - scaling - security - observability & automation - cost - troubleshooting - real-world patterns

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Executive Summary

A NAT (Network Address Translation) Gateway lets instances in private subnets make outbound connections to the internet - for OS updates, package installs, and third-party API calls - while preventing the internet from initiating connections back to them. It is a managed, AZ-scoped service that performs source NAT using an Elastic IP and sits in a public subnet. NAT Gateways are a frequent source of cost surprises and AZ-failure blind spots, making them a favorite senior-interview topic. This guide covers internals, packet flow, failure modes, scaling, security, observability, automation, cost, and troubleshooting.

1. The Problem It Solves

Private instances must stay unreachable from the internet for security, yet still need to pull updates and call external services. A NAT Gateway squares that circle: outbound allowed, inbound (unsolicited) blocked.

Core needs the NAT Gateway addresses

- **Safe egress:** Outbound internet for private hosts without giving them public IPs.
- **Inbound protection:** Drop unsolicited inbound connections by design (stateful translation).
- **Managed scale:** Handle high connection volume without managing NAT instances.
- **Stable source IP:** Present a consistent Elastic IP to external services for allow-listing.

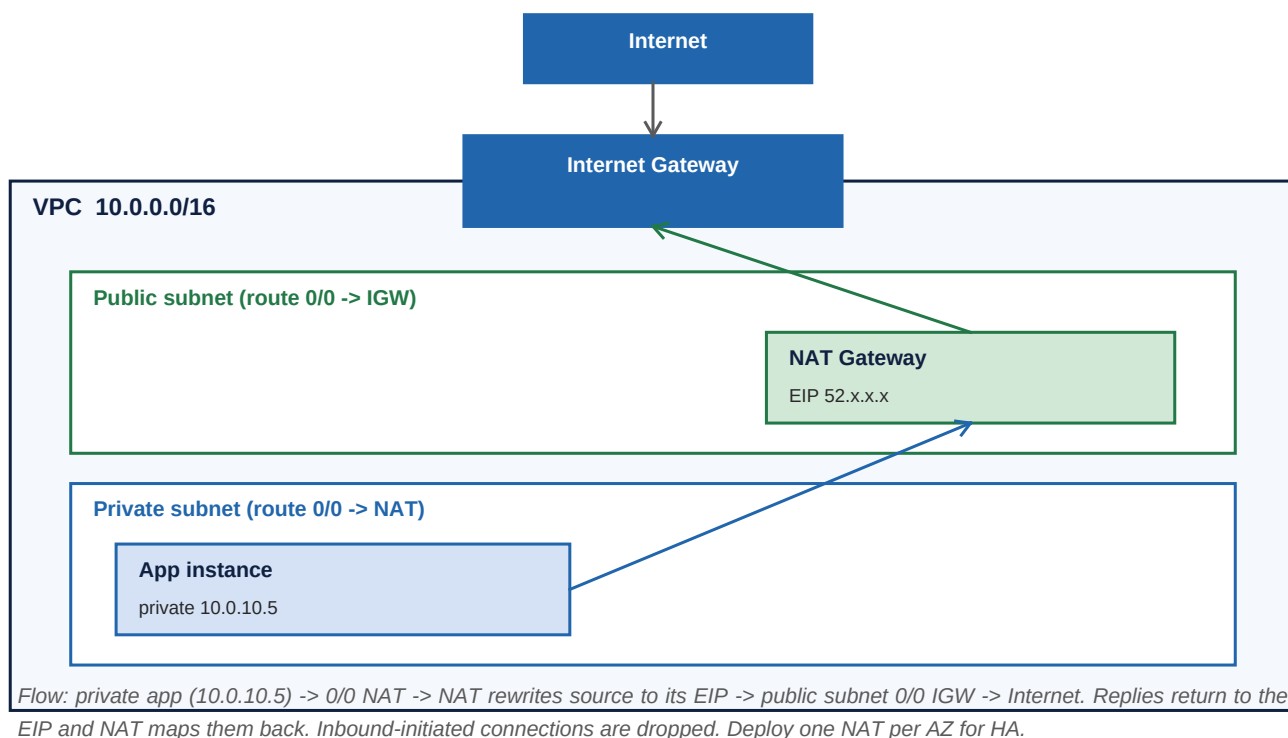
2. How It Works Internally

A NAT Gateway is deployed into a specific public subnet (and thus a specific AZ) and is given an Elastic IP. Private subnets route 0.0.0.0/0 to the NAT Gateway. It maintains a stateful translation table: it rewrites the private source IP/port to its own EIP and a chosen source port, tracks the flow, and reverses the translation for return packets.

- It supports a large number of simultaneous connections and scales bandwidth automatically up to tens of Gbps.
- Source-port space per unique destination is finite (~64k per destination IP:port), so very high fan-out to ONE endpoint can cause port exhaustion.
- It is AZ-scoped: a NAT Gateway only survives as long as its AZ; you need one per AZ for HA.
- It is stateful, so only the responses to outbound flows are allowed back in.

Reference Diagram - NAT Gateway data path

A NAT Gateway lets instances in private subnets initiate OUTBOUND internet connections (updates, API calls) while blocking unsolicited INBOUND connections. It performs source NAT using its Elastic IP and lives in a public subnet.



3. Traffic Flow, Packet Flow & Routing Decisions

Outbound via NAT

- Private instance (10.0.10.5) -> external API; SG egress evaluated.
- Private subnet route table: 0.0.0.0/0 -> nat-gw target.
- NAT GW rewrites source to its EIP + a source port; records the flow.
- Public subnet route table: 0.0.0.0/0 -> IGW; packet exits.
- Replies arrive at the EIP; NAT maps them back to 10.0.10.5; SG allows the return statefully.

AZ-affinity gotcha

Route each AZ's private subnet to the NAT Gateway IN THAT SAME AZ. If AZ-b private subnets route to a NAT in AZ-a, you pay cross-AZ data charges AND lose egress for AZ-b if AZ-a fails.

4. Advantages & Disadvantages

Advantages

- Fully managed, auto-scaling, no NAT instances to patch.
- Stateful inbound protection by default.
- Stable EIP for external allow-listing.
- High throughput and connection capacity.

Disadvantages / costs

- Charged per hour AND per GB processed - a top cost-surprise.
- AZ-scoped: needs one per AZ for true HA (more cost).
- Port exhaustion possible with massive fan-out to a single endpoint.
- Outbound-only - cannot accept inbound-initiated connections.

5. Common Failure Modes

Failure	Symptom	Typical root cause
Single-AZ NAT	AZ failure kills egress	Only one NAT GW; other AZs route to it.
Wrong route	No outbound	Private subnet 0/0 not pointing at NAT (or points at IGW with no public IP).
Port exhaustion	Intermittent connect errors	Huge fan-out to one dest IP:port exhausts source ports.
NAT in private subnet	NAT itself can't egress	NAT GW placed in a subnet without 0/0 -> IGW.
Bandwidth/conn limits	Throttling at extreme scale	Exceeding per-GW limits; need multiple GWs.
Cost blowout	Surprise bill	All egress (incl. S3) routed through NAT instead of endpoints.

6. Scaling Considerations

- **One NAT per AZ:** Deploy a NAT Gateway in each AZ and route locally for HA and to avoid cross-AZ charges.
- **Endpoints to offload:** Use gateway/interface endpoints so S3/DynamoDB and AWS APIs bypass NAT entirely.
- **Spread destinations:** Port exhaustion is per dest IP:port - many destinations scale fine; single-endpoint fan-out does not.
- **Multiple NATs:** For extreme egress, distribute across several NAT GWs.

7. Security Implications

- NAT prevents inbound-initiated connections - good default for private hosts.
- It does NOT filter outbound destinations; add egress filtering (Network Firewall / proxy) if needed.
- Its EIP can be allow-listed by partners; protect that IP's stability.
- Monitor for data-exfiltration patterns in egress traffic.

8. Observability, Reliability & Automation

Observability

- CloudWatch metrics: BytesOutToDestination, ActiveConnectionCount, ErrorPortAllocation (port exhaustion signal), PacketsDropCount.
- Flow Logs to attribute egress to source instances.

Reliability

- Per-AZ NAT Gateways with per-AZ routing remove the cross-AZ SPOF.
- Alarm on ErrorPortAllocation and packet drops.

Automation

- Provision one NAT per AZ via IaC, wiring per-AZ route tables.
- Cost guardrails: detect S3/DynamoDB traffic flowing through NAT.

9. Cost Considerations

Cost driver	Why it adds up	Mitigation
Hourly charge	Each NAT GW bills per hour, per AZ.	Right-size AZ count; tear down dev/test NATs.
Data processing	Per-GB processed on top of transfer-out.	Route S3/DynamoDB/AWS APIs via VPC endpoints (no NAT).
Cross-AZ via NAT	Routing to a NAT in another AZ adds transfer cost.	Per-AZ NAT + per-AZ routing.

10. Troubleshooting Approach

For 'private instance can't reach the internet':

- 1. Private subnet route table: 0.0.0.0/0 -> the correct (same-AZ) NAT GW?
- 2. Is the NAT GW in a PUBLIC subnet whose table has 0/0 -> IGW?
- 3. Does the NAT GW have an Elastic IP and 'available' status?
- 4. SG egress + NACL (both directions, ephemeral ports) allow it?
- 5. Check ErrorPortAllocation for exhaustion; read Flow Logs.

Mnemonic

'NAT route -> NAT in public subnet -> EIP -> SG/NACL.' Remember the NAT itself needs an IGW path to actually egress.

11. Real-World Examples & Patterns

- **Per-AZ NAT:** One NAT GW per AZ; each private subnet routes to its local NAT.
- **Endpoint offload:** Gateway endpoints for S3/DynamoDB + interface endpoints for AWS APIs to cut NAT cost.
- **Centralized egress:** Route all spoke-VPC egress through NAT in a shared egress VPC via Transit Gateway.
- **Allow-listed source:** Partner allow-lists the NAT's EIP for outbound API calls.

12. Senior-Level Trade-offs & Talking Points

- NAT Gateway vs self-managed NAT instance: managed HA/scale vs cost control and custom features.
- Per-AZ NAT (HA, no cross-AZ fees) vs single NAT (cheaper, risky).
- Centralized egress VPC (governance, fewer NATs) vs per-VPC NAT (lower latency, blast-radius isolation).
- VPC endpoints reduce NAT cost but add their own per-endpoint charges - model the break-even.

Whiteboard checklist

(1) Draw private subnet -> NAT (public subnet) -> IGW. (2) Show source NAT to EIP. (3) One NAT per AZ with local routing. (4) Add endpoints to offload S3. (5) Discuss cost (hourly + per-GB), port exhaustion, and AZ failure.

13. Quick Glossary

Term	Meaning
NAT	Network Address Translation - rewrites source IP/port for egress.
Source NAT	Translating the source address of outbound packets.
EIP	Elastic IP attached to the NAT for a stable egress address.
Port exhaustion	Running out of source ports to ONE destination.
AZ-scoped	A NAT GW lives in one AZ; needs peers for HA.
Gateway endpoint	Free private path to S3/DynamoDB, bypassing NAT.
ErrorPortAllocation	CloudWatch metric signaling port exhaustion.